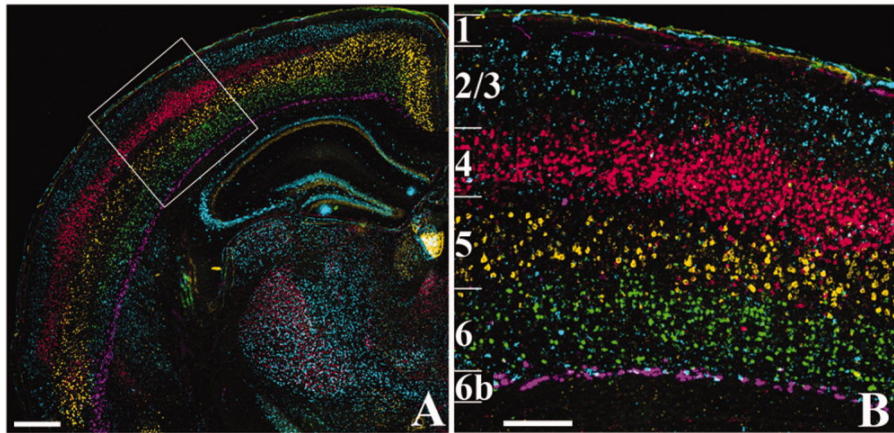




Boyle et al. (2011)

The cortical microcircuit

Layers of cortex

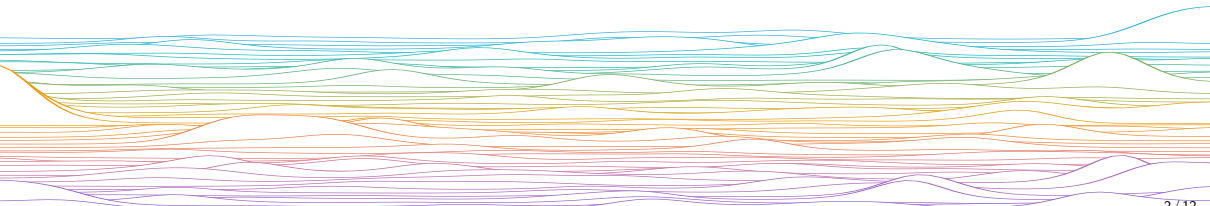
Computational Neuroscience

University of Bristol

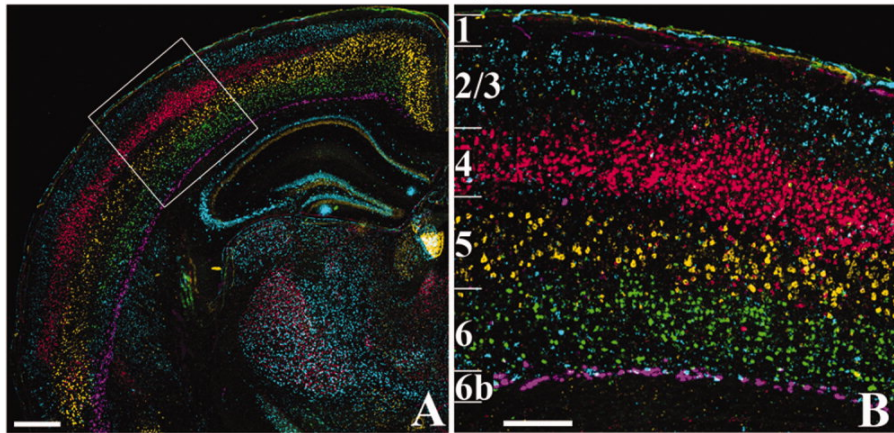
mrule

Learning outcomes:

- ▶ The cortical microcircuit
 - How many layers are (typically) in neocortex?
 - Which is the main layer receiving thalamic input?
 - Which layers give rise to cortico-cortical connections?
 - Makeup of inhibitory, excitatory cell types?



The cortical microcircuit

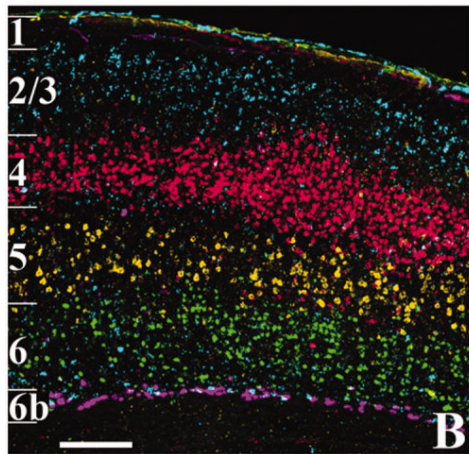


Boyle et al. (2011)

Onions have Cortex has layers

6-layer neocortex

- 1 (top, mostly neurites)
- 2 Receives cortico-cortical connections
- V1 complex cells
- 3 Sends cortico-cortical connections V1 complex cells
- 4 Gets thalamic input (e.g. sensory) V1 simple cells
- 5 sends signals out of the brain (e.g. motor)
- 6 (deepest) feedback to thalamus

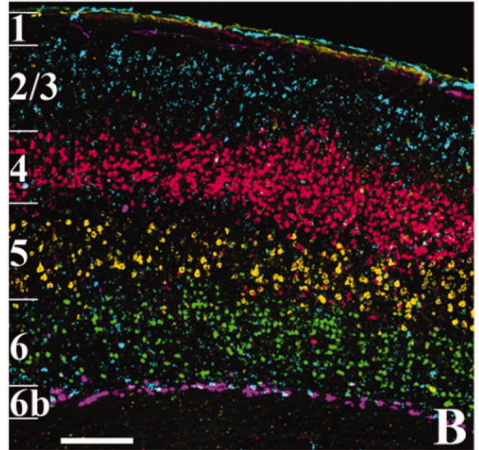


Boyle et al. (2011)

Onions have Cortex has layers

Parts of the brain that evolved earlier have fewer layers

- ▶ 3-layer archicortex (123)
 - Hippocampus: Dentate Gyrus, CA3, CA1
- ▶ 3–4-layer paleocortex (not covered) (1235)
- ▶ 3–4-layer periarthicortex (1235)
 - Subiculum
- ▶ 5–6-layer mesocortex (123456)
 - Entorhinal Cortex



Boyle et al. (2011)

The cortical microcircuit

The mammalian neocortex is fairly homogeneous: most parts of it look similar (with important exceptions). Some aspects of V1 architecture apply to cortex in general

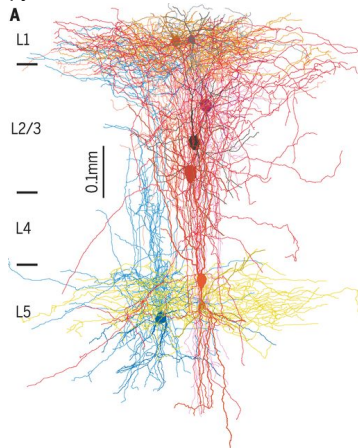
Each layer has a mix of distinct excitatory and inhibitory cell types

- ▶ More excitatory neurons than inhibitory neurons
- ▶ $\approx 80/20\%$ E/I split
- ▶ More types of inhibitory (10–20) than excitatory (4–6)

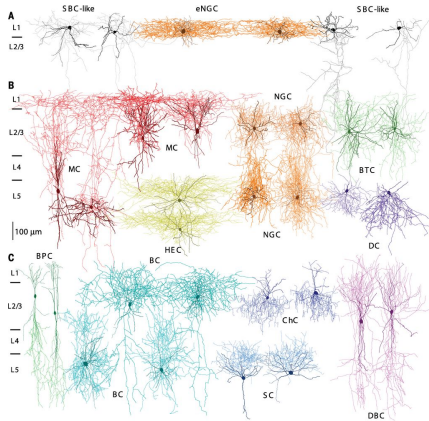
Information flow:

- ▶ Thalamus $\rightarrow$ layer 4 $\rightarrow$ layer 2/3 $\rightarrow$ other parts of the brain and layer 5
- ▶ Layer 5: outputs (usually motor-like)
- ▶ Layer 6: Feedback modulation of cortico-thalamic loops

Excitatory (glutamatergic) pyramidal cells

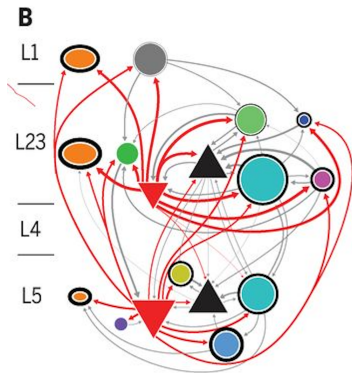


Inhibitory (GABAergic) interneurons



Jiang et al. (2015)

Circuit Diagram



pause

6-layered neocortex (oversimplified)

- ▶ Layer 4: _____ from thalamus, e.g. retinal signals arrive in V1 via LGN
- ▶ Layer 2/3: Local recurrent processing, sends signals within _____
- ▶ Layer 5: Sends motor-like or motor-related outputs
- ▶ Layer 6: Closes thalamic feedback loop

V1 tuning curves

- ▶ Simple cells
 - Found in _____ that receive inputs from the LGN (part of the thalamus)
- ▶ Complex cells
 - Found in _____

6-layered neocortex (oversimplified)

- ▶ Layer 4: Input from thalamus, e.g. retinal signals arrive in V1 via LGN
- ▶ Layer 2/3: Local recurrent processing, sends signals within cortex
- ▶ Layer 5: Sends motor-like or motor-related outputs
- ▶ Layer 6: Closes thalamic feedback loop

V1 tuning curves

- ▶ Simple cells
 - Found in V1 layers 4 and 6 that receive inputs from the LGN (part of the thalamus)
- ▶ Complex cells
 - Found in V1 layers 2/3 and 6, and higher-order visual areas

end

